Chapter 6 : Blazor

Er. Piyush Pant

Blazor

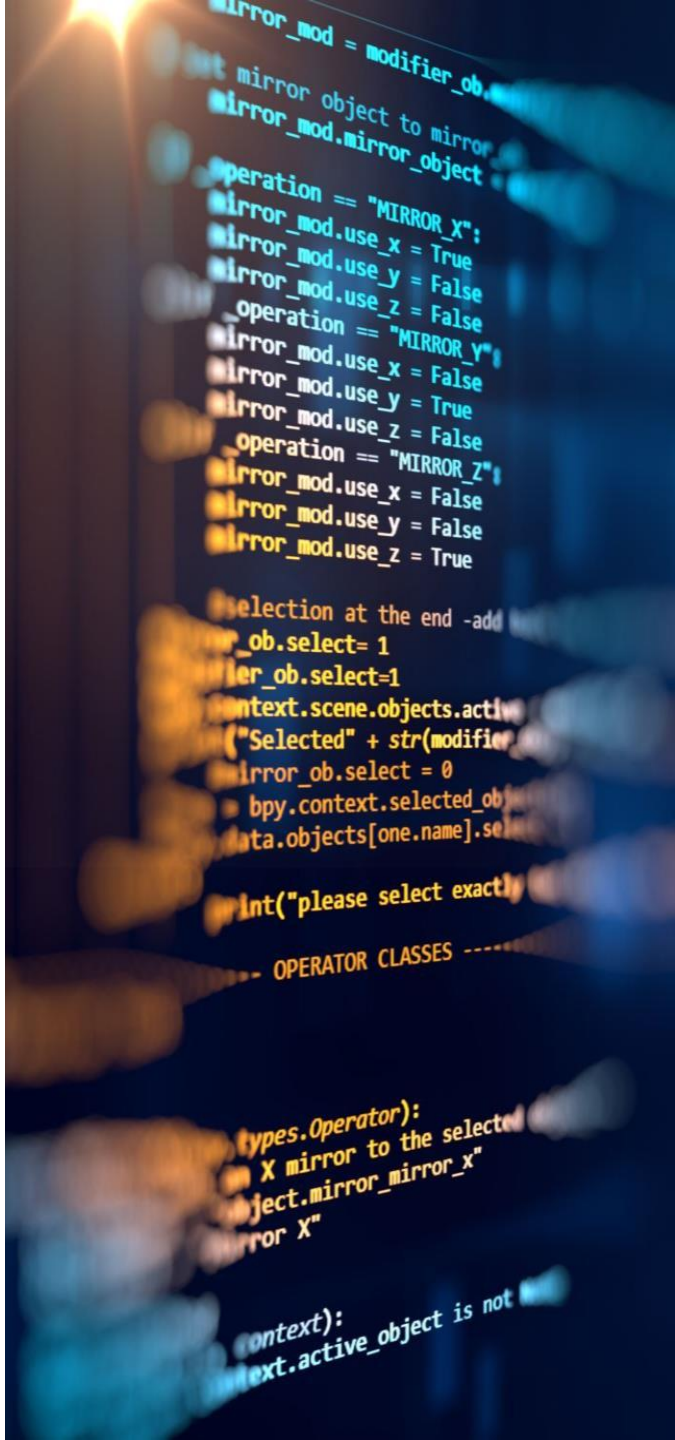
- Web development has traditionally required two languages: **C#** (or any backend language) for server-side logic and **JavaScript** for client-side interactivity. But what if you could build full-stack web applications using just C#?
- That's where Blazor comes in. Blazor is a modern framework from Microsoft that allows developers to build interactive web UIs using C#, Razor, and .NET without writing JavaScript for most scenarios.

Blazor

- Blazor is a Microsoft web framework that lets developers build interactive web applications using C# and .NET instead of JavaScript. It’s powerful for .NET developers but comes with trade-offs in performance, ecosystem maturity, and browser support.

Advantages of Blazor

- **C# for front-end**: Developers can use existing C# skills instead of learning JavaScript frameworks like React or Angular.
- **Code reusability**: Shared libraries and logic across client and server.
- **Integration with .NET ecosystem**: Seamless use of NuGet packages, Entity Framework, and other .NET tools.
- **Strong tooling**: Visual Studio and Azure integration provide a robust development environment.
- **Cross-platform**: Works on Windows, Linux, and macOS.
- **Rapid development**: Familiar syntax and reduced context switching between languages.
- **Future-proof**: Backed by Microsoft, with ongoing improvements and enterprise adoption.



Disadvantages of Blazor

- **Performance limitations**: Blazor WebAssembly apps can be slower than JavaScript frameworks, especially on initial load.
- **Large payload size**: WASM apps often have bigger download sizes, affecting load times.
- **Smaller ecosystem**: Compared to React, Angular, or Vue, Blazor has fewer third-party libraries and community resources.
- **Browser compatibility**: While modern browsers support WebAssembly, older ones may not.
- **Server dependency (Blazor Server)**: Requires constant connection; latency and scalability can be issues.
- **Learning curve for non-.NET developers**: Developers outside the Microsoft ecosystem may find it harder to adopt

Blazor Hosting Models

Blazor Server

Runs on the server; UI updates are sent to the browser via **SignalR**.

Advantages:

- Small initial download size (fast startup).
- Full access to server resources (e.g., databases, APIs).
- Centralized security and easier updates.

- **Disadvantages**:
  - Requires constant network connection.
  - Latency can affect responsiveness.
  - Scalability limited by server load.

  Best for: Enterprise/internal apps where users have reliable connections.

Blazor Hosting Models

Blazor WebAssembly (WASM)

Runs directly in the browser using **WebAssembly**;

no server dependency for UI logic.

Advantages:

- Works offline once loaded.
- Scales better since computation is client-side.
- No need for continuous server connection.

Disadvantages:

- Larger initial download (runtime + app).
- Slower startup compared to Blazor Server.
- Limited access to server resources unless APIs are exposed. **Best for**: Public-facing apps where offline support and scalability matter.

Blazor Hosting Models

Blazor Hybrid

Combines Blazor with native frameworks like **.NET MAUI** or **Electron** to run on desktop or mobile.

Advantages:

- Access to native device APIs (camera, sensors, etc.).
- Single codebase for web, desktop, and mobile.
- Rich UI with native performance.

Disadvantages:

- Larger app size.
- More complex deployment.
- Requires knowledge of native platforms.

Best for: Cross-platform apps needing both web and native features.

Feature	Blazor Server	Blazor WebAssembly	Blazor Hybrid
Execution	Server via SignalR	Browser via WebAssembly	Native desktop/mobile + Blazor
Startup Speed	Fast (small payload)	Slower (large payload)	Moderate (depends on platform)
Offline Support	❌ No	✅ Yes	✅ Yes
Scalability	Limited by server resources	Client-side scaling	Depends on native platform
Best Use Case	Enterprise apps, intranets	Public apps, offline scenarios	Cross-platform native apps

Blazor One way Binding

Data flows only from C# → UI.

The UI shows the value of a property, but user changes don't automatically update it.

```
@page "/oneway"
<h3>One-Way Binding Example</h3>
<p>Message: @message</p>
<button @onclick="ChangeMessage">Change Message</button>
@code {
    private string message = "Hello from Blazor!";
    private void ChangeMessage()
    {
        message = "Message updated!";
    }
}
```

Here, `@message` is displayed in the UI.

Clicking the button updates the property in C#, and the UI reflects it.

But if the user typed something in the UI, it wouldn't change `message` automatically.

Blazor Two Way Binding

Data flows both ways: C# ↔ UI.

The UI updates when the property changes, and the property updates when the user changes the UI (e.g., typing in an input box).

```
@page "/twoway"
<h3>Two-Way Binding Example</h3>
<p>Enter your name:</p>
<input @bind="name" />
<p>Hello, @name!</p>
@code {
    private string name = "Blazor User";
}
```

Here, typing in the input box updates the `name` property automatically, and the greeting updates in real time.